



MIME: Machine Intelligence for Motion Exegesis

Studying Historical Styles of Theatrical Pose and Gesture with AI

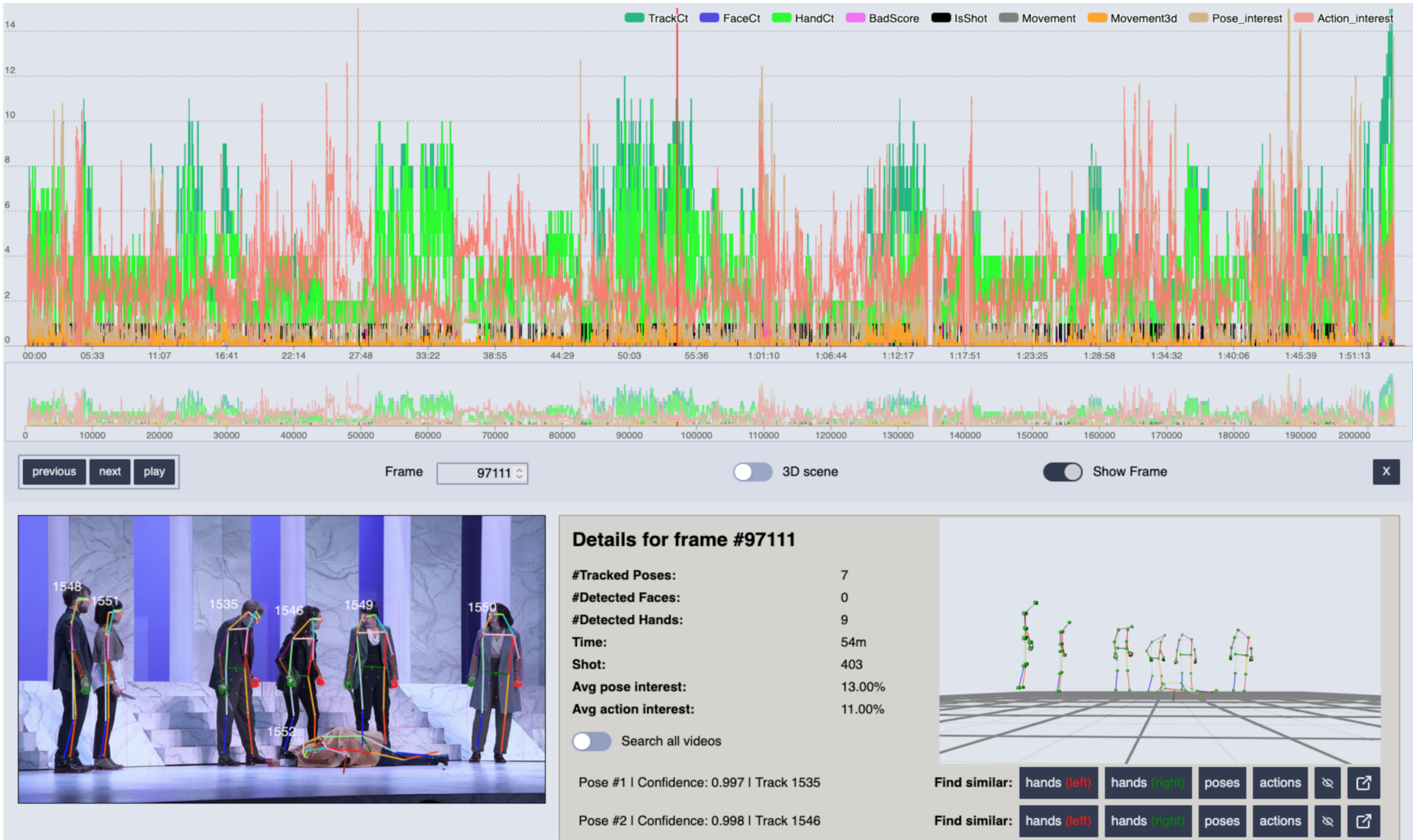


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Pose and gesture stylometry: how do AI-powered analyses of performances relate to conventional wisdom regarding the signature choreographic styles of "auteur" directors, or how famous actors embody roles?

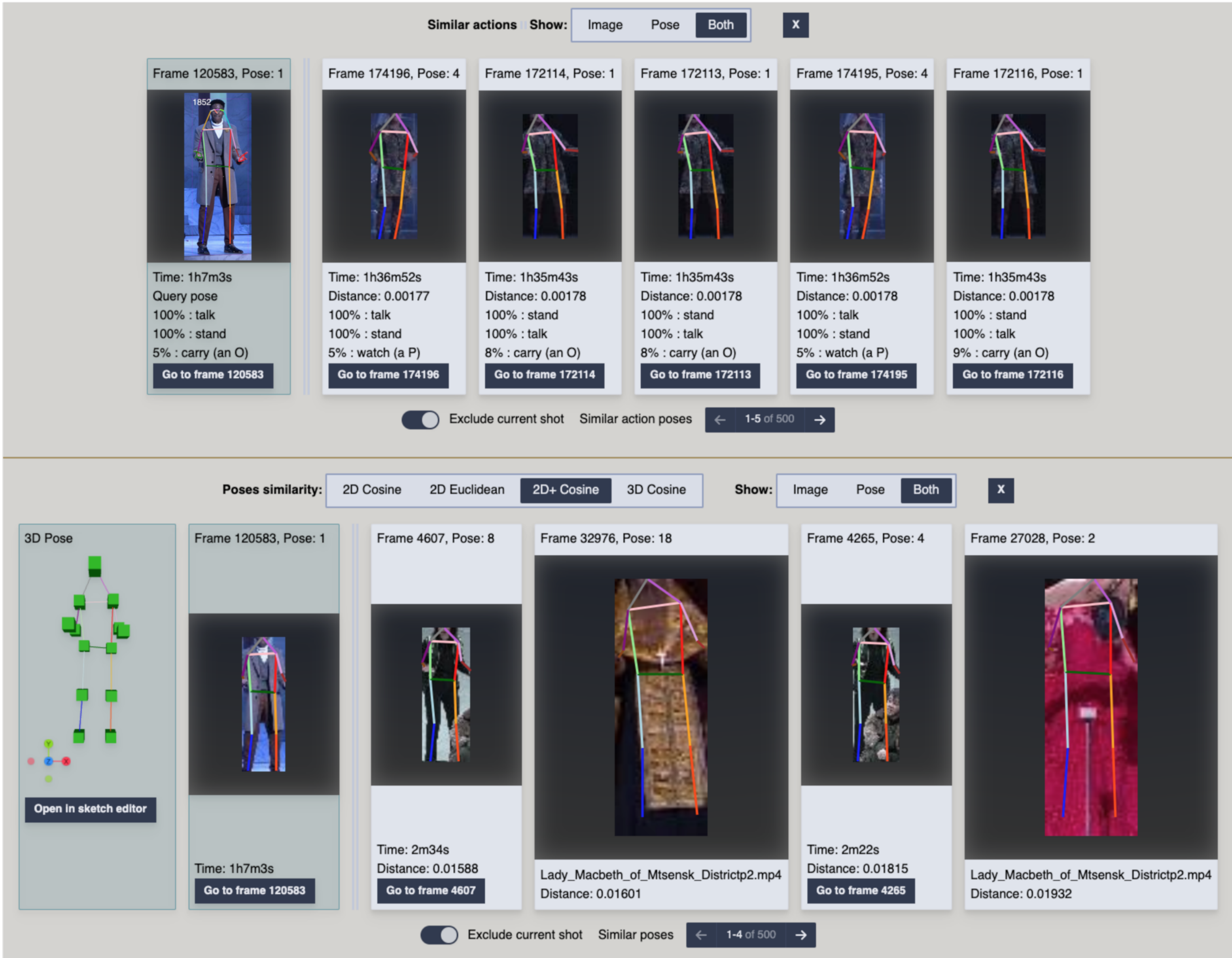
Historical influences: can we use theories of acting, dance and oratory from earlier eras to navigate the latent semantic spaces of pose and gesture, and possibly trace influences to the present day?



MIME's timeline view, displaying pose and motion data for a performance, with details and 3D body armatures for a selected frame.

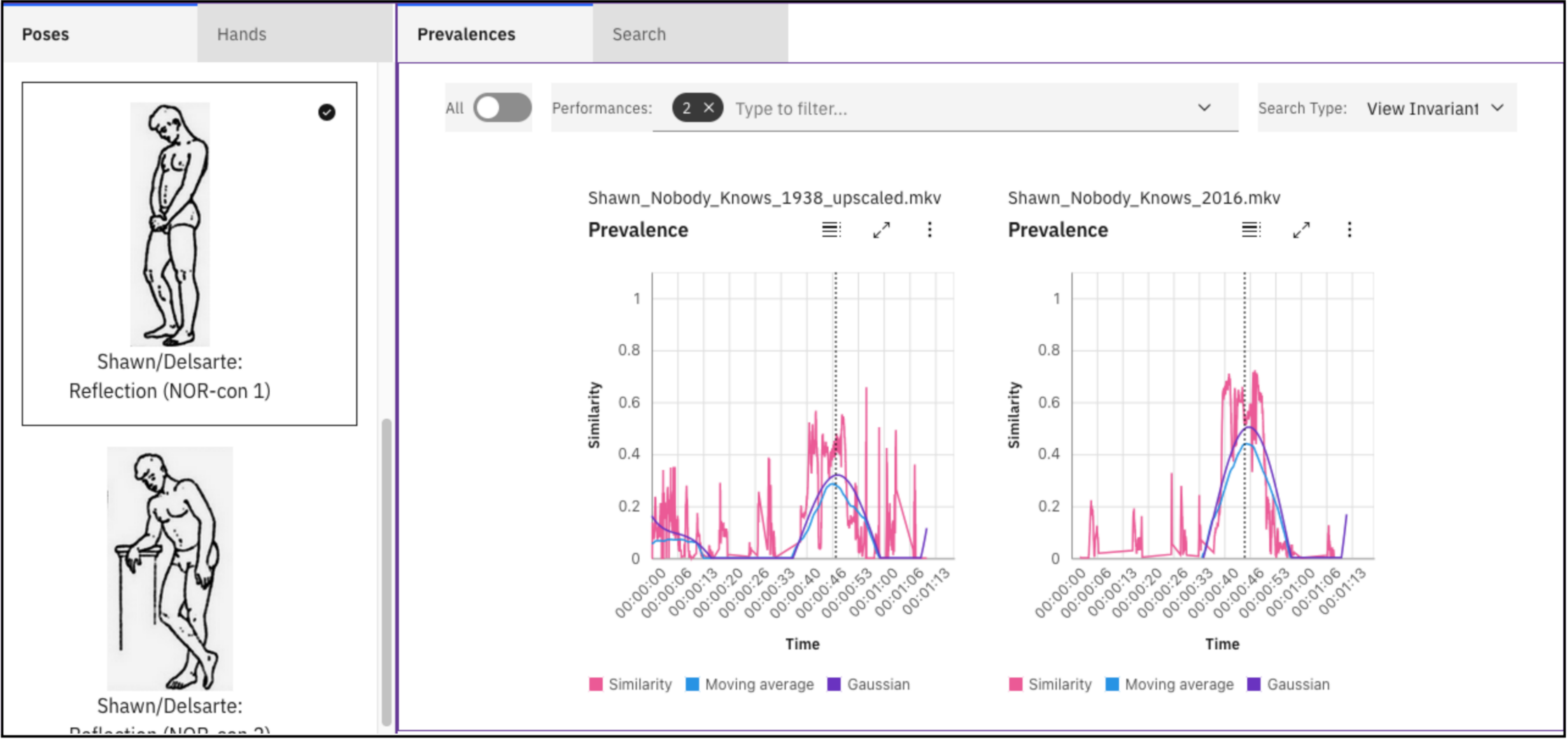
Core materials: we work with "in-the-wild" recordings (from regular cameras) of contemporary and historical theater performances. Our studies so far focus on a few dozen performances of 1-3 hours each, but there is potential to scale up to thousands of performances.

Key methods: we apply several AI models based on vision transformers to detect body poses, hand gestures and facial landmarks and to recognize actions, faces/costumes, and characteristic poses despite varying camera angles. Models for **3D depth estimation** from 2D video and **person tracking** over time in congested environments have been essential to our progress.

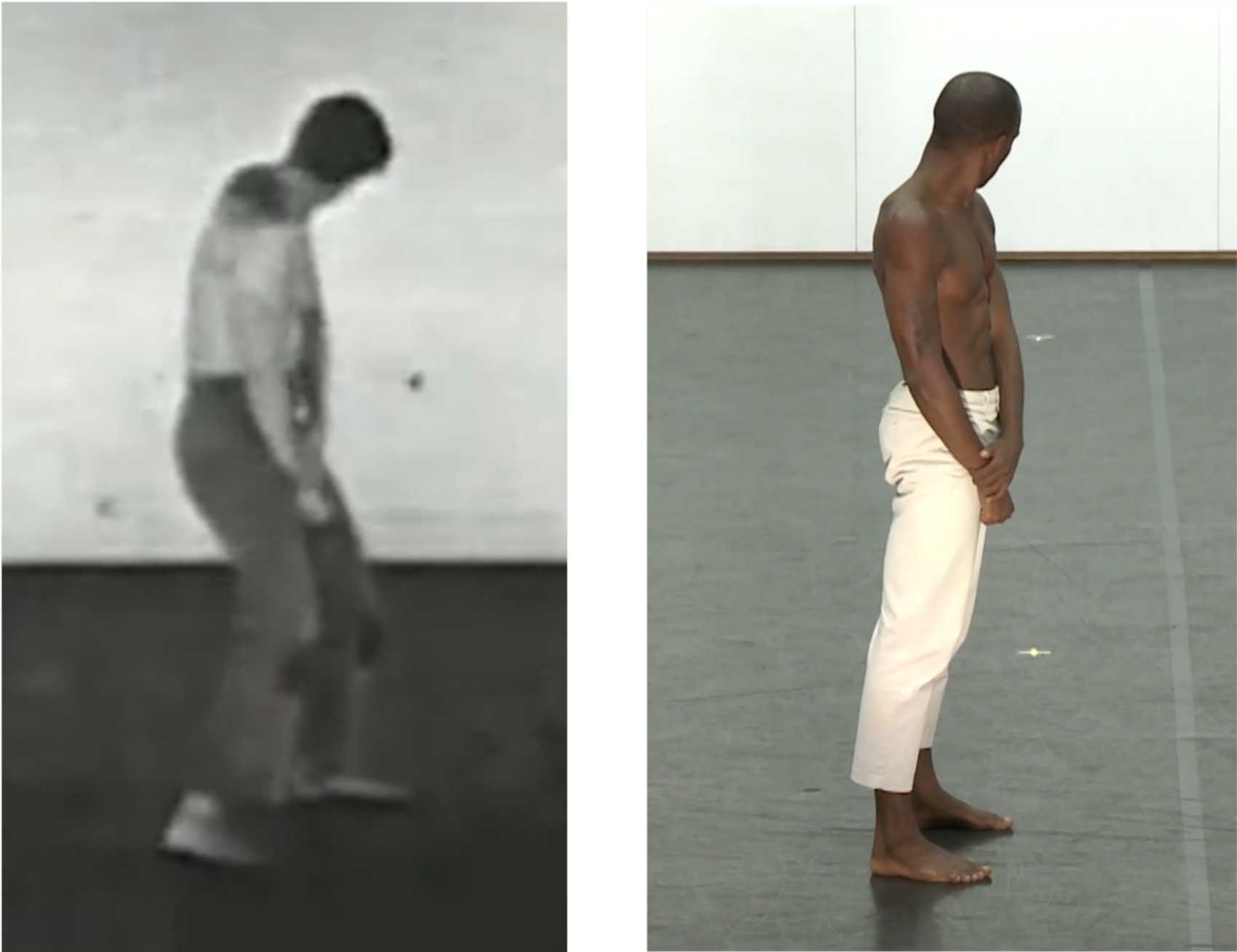
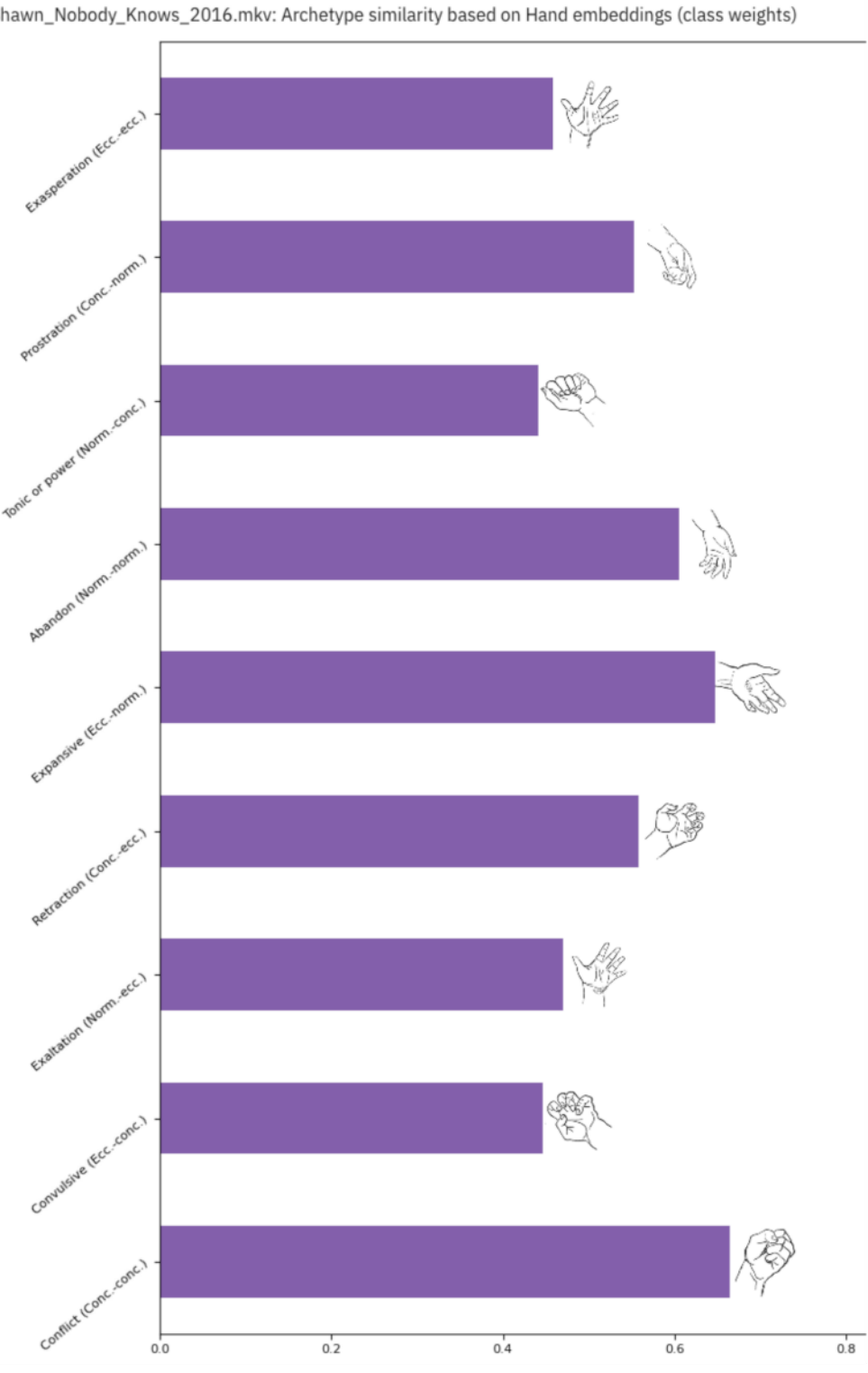


Top: the MIME interface displaying the results of an action similarity search across 31 recorded performances, using 60-element action embedding vectors for each frame of each tracked pose, which are distilled into three actions from the Atomic Visual Actions (AVA) ontology.

Bottom: the results of a pose similarity search across the 31 recorded performances, using 16-element view-invariant pose embedding vectors for each tracked pose, indexed for approximate nearest neighbor search based on cosine similarity.



Pose prevalence time-series plots based on cosine similarity of the detected poses to the "Reflection" pose archetype from the system of François Delsarte (1811-1871), as illustrated by American modern dance luminary Ted Shawn (1891-1972), in two performances of "Nobody Knows the Trouble I've Seen" from Shawn's *Four Dances Based on American Folk Music* (1931).



Left: prevalences of François Delsarte's nine hand gesture archetypes, quantified as the average cosine similarity of all detected hand poses to each archetype, in "Nobody Knows" (2016). Right: extracted frames from the times indicated in the time-series plots above; the left is from Shawn's own performance in 1938, and the right is from Davon Rainey's performance in 2016, directed by Adam H. Weinert.

Collaboration opportunities

- Working with experts in developing and running **large multimodal models** (also taking into account text, audio and object detection), perhaps finding ways to incorporate **historical paradigms** of pose and gesture into such models.
- Consulting with specialists in **statistical feature extraction** and processing of **large quantities of data** derived from video sources, as each frame can generate thousands of data points.
- Applying emergent practices involving the **exploration of semantic "embedding" spaces** to navigate the landscape of poses and actions associated with a director, actor or other grouping of performances, perhaps also **using historical archetypes as "signposts"** within those spaces.

Challenges

- **Characterizing gestures** (as variable sequences of hand and body poses that unfold over time) so they can be searched and compared remains much more difficult than just analyzing static poses.
- **Identifying "signature" poses and gestures** requires sophisticated statistical methods and/or deep-learning models that need to be customized and trained; so far we've mostly used unmodified, "out-of-the-box" models.
- Models still struggle with recognizing faces, bodies and hands from many **angles, distances and lighting conditions**, which constrains our work.

References

1. Rajasegaran, Jathushan, Georgios Pavlakos, Angjoo Kanazawa, Christoph Feichtenhofer and Jitendra Malik. "Tracking People by Predicting 3D Appearance, Location & Pose." In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
2. Shawn, Ted. *Every Little Movement: A Book about François Delsarte, the Man and His Philosophy, His Science and Applied Aesthetics, the Application of This Science to the Art of the Dance, the Influence of Delsarte on American Dance*. Brooklyn: Dance Horizons, 1968.
3. Skybetter, Sydney. "The Haunting of Adam Winert by Ted Shawn." *Jacob's Pillow Dance Interactive*, January 2019.



MIME system technical architecture

